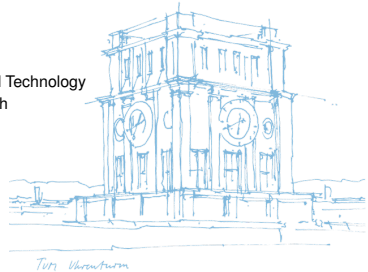


An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

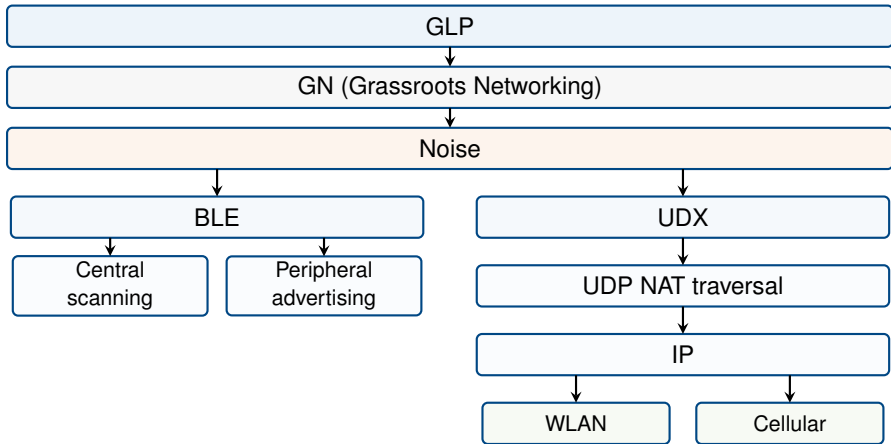
Dan Bachar, David Guzman

Monday 18th May, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

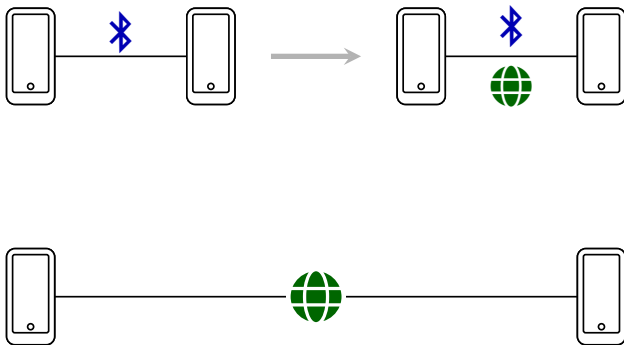


Architecture



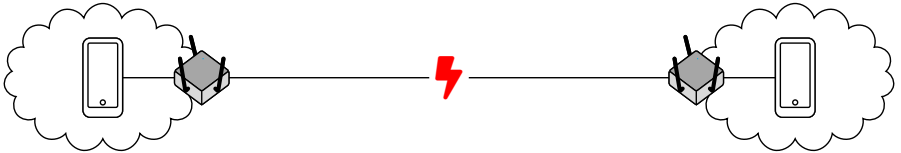
- GN uses Noise to establish secure sessions over BLE or UDX.
- BLE operates through central scanning and peripheral advertising roles.
- UDX uses UDP for NAT traversal over IP, carried by WLAN or cellular.

Dual-stack comms

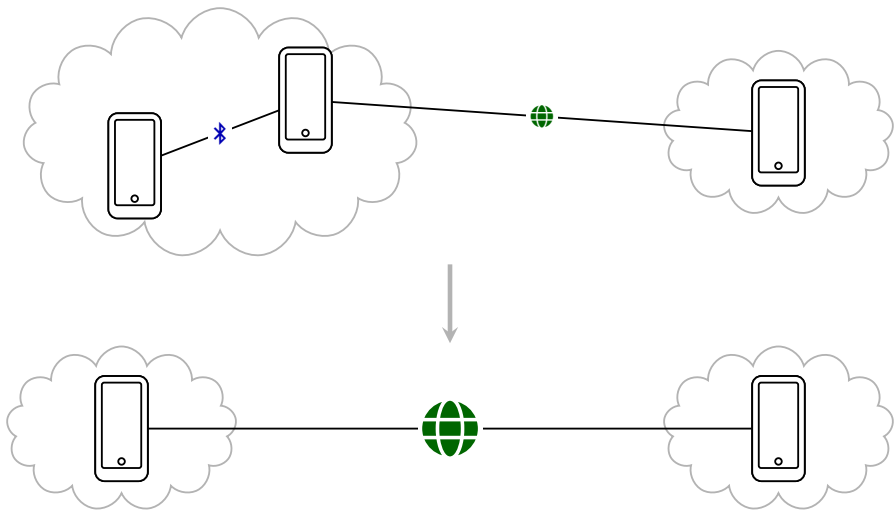


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

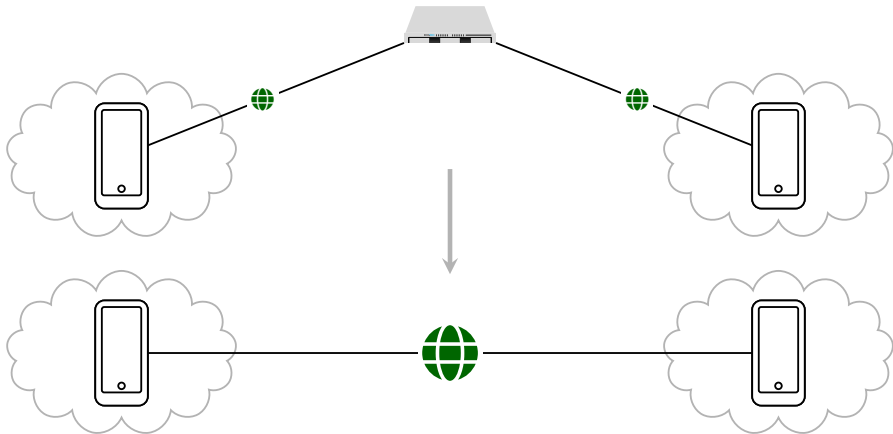
NAT



Well-connected friends



Rendezvous Server



Since last weekly

- Migrated BLE communication layer from two separate central/peripheral plugins to one `grassroots_bluetooth_layer`: unified event model, shared discovery UUID, post-connect ANNOUNCE identity, and BLE role toggle.
- Reworked UDX into a dual-stack Internet transport: IPv4/IPv6 sockets, multi-candidate addresses, live network/public-address refresh, and best candidate-pair selection.
- Aligned rendezvous/NAT traversal with GLP Networking API doc: two-sided punch, RV fan-out (A and B try each others' RV servers when the connection breaks), friends-of-friends mediation (friends can facilitate rendezvous for mutual friends).
- Persist friend data in encrypted local storage so that connections can be initiated post restart.
- Added Noise XX handshake for secure channel encryption: messaging and signaling with Noise XX across clients and rendezvous anchors.
- Hardened delivery over UDX/BLE: ACKs after fragment reassembly, increase max UDX payload size for photos, and stale-session/socket recovery (app loses connections when backgrounded)

Mobility model: city-scale BLE relay

Protocol question

- Given real city encounters, when does sealed BLE relay deliver?
- Compare direct, friend-only, and open relay policies.
- Existing mobility traces give contacts; we measure protocol outcomes.

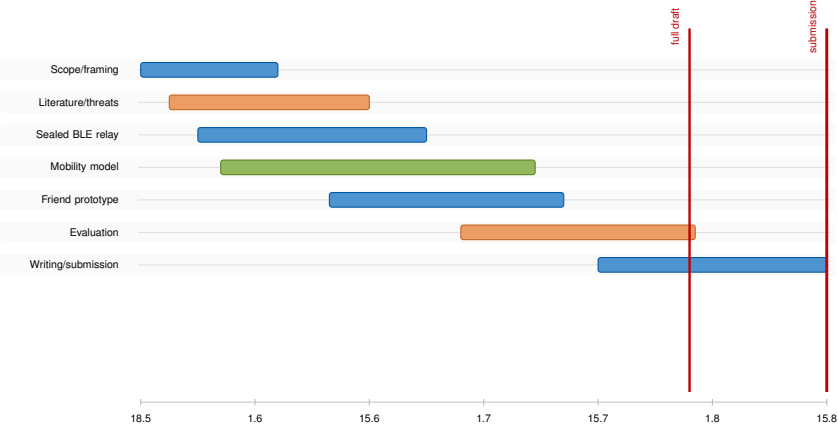
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost/risk: relay bytes, duplicates, queues, scan/advertise time, battery drain.

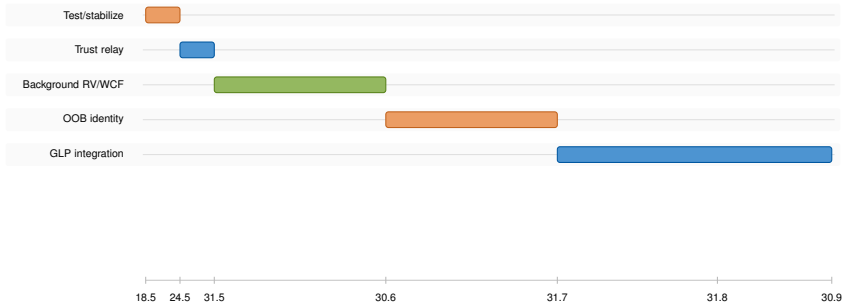
How

- Use existing BLE/contact models as baselines or priors.
- Calibrate with encrypted local trace logs from our Flutter stack.
- Replay traces in simulator with relay policy, TTL, ACKs, and queues.

City-scale BLE thesis



Internet-scale UDX implementation



- **Test/stabilize:** run multiple devices across Wi-Fi, cellular, IPv4/IPv6, NATs, roaming, foreground/background transitions, and UDX connection migration.
- **Trust relay:** implement relay modes: open relay, friends-only relay, and GPS-defined trusted zones.
- **Background RV/WCF:** keep rendezvous anchors and well-connected friends useful while the app is backgrounded.

- **OOB identity:** peer links for identity exchange, intent binding, and bootstrapping friendship/connection state.
- **GLP integration:** expose the stabilized transport and rendezvous behavior through the GLP Networking API.

Internet-scale UDX: possible extensions

Transport and discovery

- **BLE discovery optimization:** energy-aware scan/advertise duty cycling without losing too many encounters.
- **Dual-transport selection:** choose BLE or UDX per message based on reachability, latency, trust, cost, and payload size.
- **Cost-aware routing:** quantify bytes, battery, retries, relay load, and latency per delivered message.

Measurement

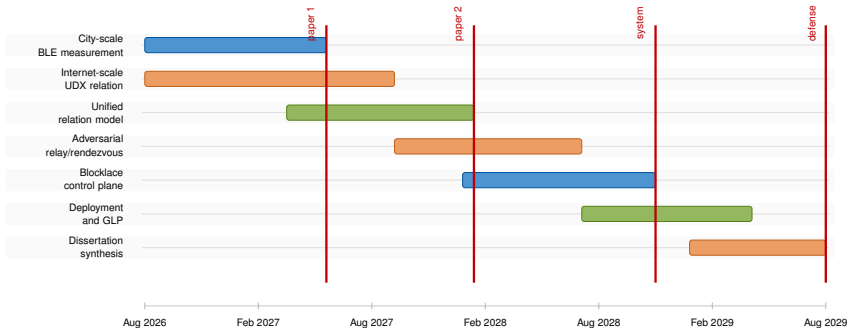
- **Well-connected friends:** characterize which peers can realistically help with signaling and rendezvous.
- **Grassroots NAT study:** measure NAT behavior, address churn, IPv4/IPv6 availability, and hole-punch success.

Trust and identity

- **Out-of-band identity exchange:** peer links should carry intent/attestation, not only a public key.
- **Friend-based relay:** safest relay mode; reduce attack surface with sealed forwarding and minimal visible metadata.
- **Geographic trust zones:** conferences, workplaces, or classrooms where local strangers may be acceptable relays.
- **Stranger relay:** highest-risk mode, but useful for emergency, protest, or disaster-connectivity scenarios.
- **Relay by trust level:** adapt forwarding policy to friend, trusted-zone, or stranger context.

Research question: once the system is stable, which relay and transport choices improve delivery enough to justify their cost and abuse surface?

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.